

SCSIT

IV Semester

Database

Exc. 5

SELECT; simple group functions

(queries for data retrieval from one DB table)

SELECT

Basic syntax for one table

SELECT [**DISTINCT**] {*, *column* [[**AS**] *alias*], ...}

FROM *table*

[**WHERE** **condition**]

[**GROUP by** group statement]

[**HAVING** groups condition]

[**ORDER by** column [ASC/DESC]]

1. Select all rows that have salary above 20000 and don't have phone

$\sigma_{salary > 20000 \text{ and } phone = 0}$ (work_place)

2. Select all rows that have laptop, phone and vehicle, but salary is below 10000 or sector is Administration.

$\sigma_{(salary < 10000 \text{ or } sector = 'Administration') \text{ and } laptop = 1 \text{ and } phone = 1 \text{ and } vehicle = 1}$ (work_place)

3. Get titles for work places that have salary below 25000 and have laptop or phone

$\Pi_{\text{title}} (\sigma_{\text{salary} \leq 25000 \text{ and } (\text{laptop}=1 \text{ or } \text{phone}=1)} (\text{work_place}))$

4. Get salaries and titles for work places in IT Sector. Rename the salary column into IT_salaries. The results should be sorted by salary amount (from smallest to biggest)

$\rho_{\text{salary/it_salaries}} (\Pi_{\text{salary,title}} (\sigma_{\text{sector}=\text{"IT"}} (\text{work_place})))$

5. Get titles and salaries for work places that have salary which is at least twice bigger than the smallest one

```
SELECT Salary, title  
FROM Work_place  
WHERE Salary >= 2 * (SELECT min(Salary) FROM Work_place)
```

GROUP functions

Count

SUM

AVG

Min

Max

StDev

These functions process data from one column.
The result from these functions (usually) is only
ONE VALUE.

Ex:

T=

A	B
1	3
3	4
3	2
<i>null</i>	3

Min(A) = **1**

Max(B) = **4**

SUM(B) = **12** Avg(B) = **3**

Count(A) = **3**

Count(*) = ?

SUM(A) + Max(B) = ?

SQL queries for group functions - G

SELECT G

FROM *table*

[WHERE condition]

G= min, max, count, avg, sum...

Table= table name

condition= statement for picking data for processing

Ex:

TABLE PC_components

6. Get the number of rows in the table

```
SELECT COUNT(*)  
FROM PC_COMPONENTS
```

Ex. 2:

TABLE PC_components

7. Determine the maximal warranty given for some product

```
SELECT MAX(WARRANTY)  
FROM PC_components
```

Ex. 3:

PC_components

**8. Determine the number of
DIFFERENT product categories**

```
SELECT count( DISTINCT category)  
FROM PC_components
```

9. Determine the average product price

10. Find minimal and maximal product price

11. Find the number of warranty types

12. Find the profit that company has, by assumption that it sold one piece from each product

(Ex. If 50% of the product price is its original + 18% AVT
profit = total amount of prices – total amount of prices *
18/100 – total amount of prices * 50/100)

13. Find number of products from CPU category

```
SELECT COUNT(*)  
FROM PC_components  
WHERE category='CPU'
```

14. Find average price for product that have 12 months warranty

15. Find maximal warranty from products that in their name contain MEMO (меморија, мемории etc.)

*some table data are written with Cyrillic letters

16. Find the category with shortest name

17. Find number of products that belong to shortest name category.